Unit 2 – Register Transfer and Micro-operations (Exam Notes)

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1. Basic Definitions

- **Digital System = digital hardware modules (registers, decoders, arithmetic units, control logic) ka interconnection.**
- **Micro-operation: registers me stored data pe perform hone wali elementary operation (jaise shift, count, clear, load).**
- **Computer ka hardware 3 cheezon se define hota hai:**
    1. **Registers aur unka function**
    2. **Registers pe hone wali micro-operations ki sequence**
    3. **Control (jo in operations ko initiate karta hai)**

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2. Register Transfer Language (RTL)

- **RTL = symbolic notation jisse registers ke beech micro-operation transfer describe karte hain.**
- **Registers ko CAPITAL letters se naam dete hain: MAR (Memory Address Register), PC (Program Counter), IR (Instruction Register), R1 (processor register).**
- **Register ke bits right se left, 0 se (n-1) tak number hote hain.**
- **16-bit register ko 2 parts me baant sakte hain: L (low byte, bits 0-7) aur H (high byte, bits 8-15).**

Register Transfer Notation Symbols

Symbol	Meaning	Example
Letters	Register denote karta hai	MAR, R2
( )	Register ka ek part	R2(0-7)
← (Arrow)	Information transfer	R2 ← R1
, (Comma)	2 micro-ops same time pe	R2←R1, R1←R2

- **R2 ← R1 ka matlab: R1 ka content R2 me copy hota hai. R1 khud change nahi hota.**
- **Control condition ke saath: if(P=1) then R2←R1 ko short me likhte hain: P: R2←R1 (colon ke baad condition true hone par hi transfer hoga).**
- **Exchange operation: T: R2←R1, R1←R2 — dono registers ke contents ek clock pulse me swap ho jaate hain.**

Timing important point: Control signal P clock ke rising edge pe activate hota hai time t pe, lekin actual transfer next clock pulse (t+1) pe hota hai.

3. Bus and Memory Transfers

Common Bus System

Bahut saare registers ke beech efficient transfer ke liye ek common bus use karte hain (har bit ke liye ek common line).

2 tarike bus banane ke:

1. **Using Multiplexers**
2. **Using Tri-state (Three-state) Buffers**

(a) Bus with Multiplexers

- **k registers, har register n-bit ka → n multiplexers chahiye, har MUX size = k×1.**
- **Selection lines (S1, S0) decide karti hain konsa register bus pe aayega.**

Example (4 registers A,B,C,D, S1S0 se select):

S1 S0 | Register Selected

0 0 | A

0 1 | B

1 0 | C

1 1 | D

- **Statement: BUS←C, R1←BUS — matlab C ka content bus pe aaya, fir bus se R1 me load hua. Agar bus system pata hai to direct likh sakte: R1←C.**

(b) Bus with Three-State Buffers

- **Three-state gate ke 3 states: logic 1, logic 0, aur high-impedance (jaise open circuit — output disconnect).**
- **Control input C=1 → output normal (buffer jaisa kaam)**
- **Control input C=0 → output high-impedance (disabled)**
- **Isse multiple buffers ke output ek hi bus line pe wire kar sakte hain bina loading problem ke.**
- **Ek time pe sirf ek hi buffer active hona chahiye — ye decoder ke through control hota hai.**

Memory Transfer

- **Read operation: memory se data nikalna → DR ← M[AR] (AR = address register, DR = data register)**
- **Write operation: memory me data likhna → M[AR] ← R1**
- **M[AR] ka matlab hai memory word jiska address AR me stored hai.**

4. Types of Micro-operations (4 Types – IMPORTANT)

Type	Kaam
Register Transfer	Bas data ko ek register se dusre me move karta hai (content change nahi hota)
Arithmetic	Numeric data pe arithmetic operation (add, subtract, etc.)
Logic	Bit-wise operations (AND, OR, XOR, etc.)
Shift	Data ko left ya right shift karna

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5. Arithmetic Micro-operations

5 basic operations: Addition, Subtraction, Increment, Decrement, Shift

- **Add: R3 ← R1 + R2 (3 registers + adder circuit chahiye)**
- **Subtract: complement + addition se hoti hai: R3 ← R1 + R2' + 1 (R2' = 1's complement, +1 karne se 2's complement ban jaata hai) → R1 + 2's complement of R2 = R1 - R2**

Binary Adder vs Full Adder

- **Full Adder: 2 bits + previous carry ka sum nikalta hai.**
- **Binary Adder: kisi bhi length ke 2 binary numbers ka sum nikalta hai (multiple Full Adders chain me).**
- **n-bit binary adder = n Full Adders (carry chain se connected, C0 input carry, Cn output carry).**

Binary Adder-Subtractor

- **Ek hi circuit se Add aur Subtract dono — har Full Adder ke saath ek XOR gate lagate hain.**
- **Mode input M control karta hai:**
  - **M=0 → Adder (B XOR 0 = B, circuit karta hai A+B)**
  - **M=1 → Subtractor (B XOR 1 = B', aur Cin=1 → circuit karta hai A + 2's complement of B = A-B)**

Binary Incrementer

- **Register me 1 add karta hai (jaise 0110→0111).**
- **Half Adders cascade me connect karke banta hai.**
- **Sabse pehle (least significant) HA ka ek input hamesha logic-1 hota hai.**
- **n-bit incrementer = n Half Adders chahiye.**

Arithmetic Circuit (4-bit) — 8 Microoperations Table

Parallel adder + multiplexers milke control karte hain kya operation ho: D = A + Y + Cin

S1	S0	Cin	Y	Operation	Result
0	0	0	B	Add	D=A+B
0	0	1	B	Add with carry	D=A+B+1
0	1	0	B'	Subtract with borrow	D=A+B'
0	1	1	B'	Subtract	D=A+B'+1
1	0	0	0	Transfer A	D=A
1	0	1	0	Increment A	D=A+1
1	1	0	1	Decrement A	D=A-1
1	1	1	1	Transfer A	D=A

Yaad rakhne ka trick: S1S0=00→Add, 01→Subtract, 10→Increment, 11→Decrement.

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6. Logic Micro-operations

- **Ye register ke har bit ko individually binary variable maan ke operate karte hain (bit manipulation).**
- **2 variables se total 16 possible logic operations ban sakti hain (16 truth table combinations).**
- **Lekin practically sirf 4 basic operations use hote hain: AND, OR, XOR, Complement — inse baaki sab derive ho sakte hain.**

Important Named Operations (yaad rakho)

Function	Micro-op	Name
F=0	F←0	Clear
xy	F←A∧B	AND
x+y	F←A∨B	OR
x⊕y	F←A⊕B	Exclusive-OR
x'	F←$\bar{A}$	Complement A
(xy)'	F←$\bar{A}$∧$\bar{B}$ (NAND form)	NAND
1	F←all 1's	Set to all 1's

Hardware

- **Har bit position ke liye gates lagte hain (AND, OR, XOR, NOT) + ek 4×1 MUX jo S1,S0 se output select karta hai.**

Applications of Logic Micro-ops (Bahut Important)

Operation	Kya karta hai	Gate Used
Selective-Set	A ke bits ko 1 karta hai jaha B me 1 ho	OR
Selective-Complement	A ke bits complement karta hai jaha B me 1 ho	XOR
Selective-Clear	A ke bits ko 0 karta hai jaha B me 1 ho	AND with B'
Mask	A ke bits clear karta hai jaha B me 0 ho	AND
Insert	Pehle mask karo, fir naye value OR karo	Mask + OR
Clear	Agar A aur B same hain to result sabhi 0	XOR

Example (Selective-set, OR operation):

1010 A before

1100 B (operand)

1110 A after (jaha B=1 wahan A=1 ho gaya)

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7. Shift Micro-operations

- **Data ka serial transfer karne ke liye use hote hain.**
- **Left shift → serial input rightmost position me jaata hai.**
- **Right shift → serial input leftmost position me jaata hai.**

3 Types of Shift:

Type	Symbol	Description
Logical	shl / shr	Serial input hamesha 0 hota hai
Circular (Rotate)	cil / cir	Bits register ke ends pe circulate hote hain, koi bit lost nahi hota
Arithmetic	ashl / ashr	Signed binary number ko shift karta hai; sign bit change nahi hota

- **Arithmetic shift-left = number ko 2 se multiply karta hai**
- **Arithmetic shift-right = number ko 2 se divide karta hai**

Hardware (Combinational Shifter)

- **Multiplexers se banta hai. n data input/output → n multiplexers chahiye.**
- **Selection input S:**
    - **S=0 → shift right**

- **S=1 → shift left**

- **2 serial inputs: IL (shift-left ke liye) aur IR (shift-right ke liye)**

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8. Arithmetic Logic Shift Unit (ALU)

- **ALU = ek common combinational circuit jisme arithmetic, logic aur shift circuits combine hote hain, common selection variables ke saath.**
- **Registers directly operate nahi karte — sabhi ek common ALU use karte hain.**
- **Poora register transfer operation (source → ALU → destination) ek hi clock pulse me complete ho jaata hai.**

ALU Function Table (14 Operations) — S3S2 decide karta hai type

S3 S2	Operation Type	Selected by
00	Arithmetic (8 ops)	S1,S0,Cin
01	Logic (4 ops: AND,OR,XOR,Complement)	S1,S0 (Cin ka effect nahi, don't care)
10	Shift Right	—
11	Shift Left	—

Important point: Input carry (Cin) sirf arithmetic operations ke liye use hota hai, logic aur shift me iska koi effect nahi hota.

Quick Revision (1-minute recap)

- **RTL: Registers ke beech data transfer symbolic form me — R2←R1 (arrow=transfer, comma=parallel ops, colon=control condition)**
- **Bus: Multiple registers connect karne ka common tarika — Multiplexer based ya Three-state buffer based**
- **Memory Transfer: Read = DR←M[AR], Write = M[AR]←R1**
- **4 Types of Micro-ops: Register Transfer, Arithmetic, Logic, Shift**
- **Arithmetic: Add/Sub/Increment/Decrement — Full Adder chain se Binary Adder banta hai; Subtract = Add + 2's complement**
- **Logic: 16 possible ops, but 4 main (AND,OR,XOR,Complement) — Applications: Selective-set(OR), Selective-complement(XOR), Selective-clear/Mask(AND), Insert(Mask+OR), Clear(XOR same numbers)**
- **Shift: Logical(0 fill), Circular(rotate, no loss), Arithmetic(sign bit safe, ×2 or ÷2)**
- **ALU: Sab kuch ek circuit me combine — S3S2 decide karta hai Arithmetic/Logic/Shift, Cin sirf arithmetic me matter karta hai**